Natural Language Processing
A Soft Introduction to Ontologies and the Semantic Web

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Outline

1. About this lesson
2. Ontologies
   - Methodologies
   - Languages
   - Tools
3. The Semantic Web

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About this lesson: what is an Ontology

“The study of that which is common to all things which exist”
Aristotle, Metaphysics

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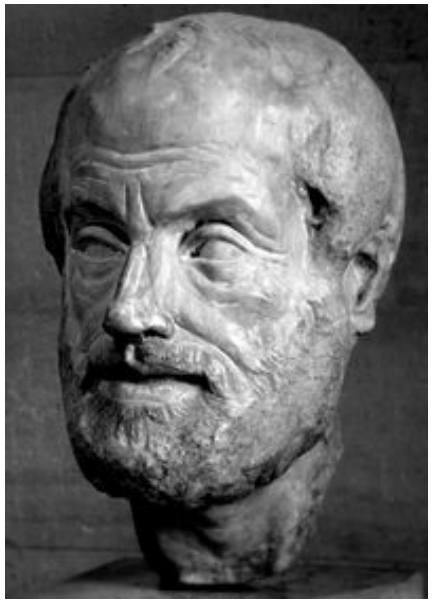
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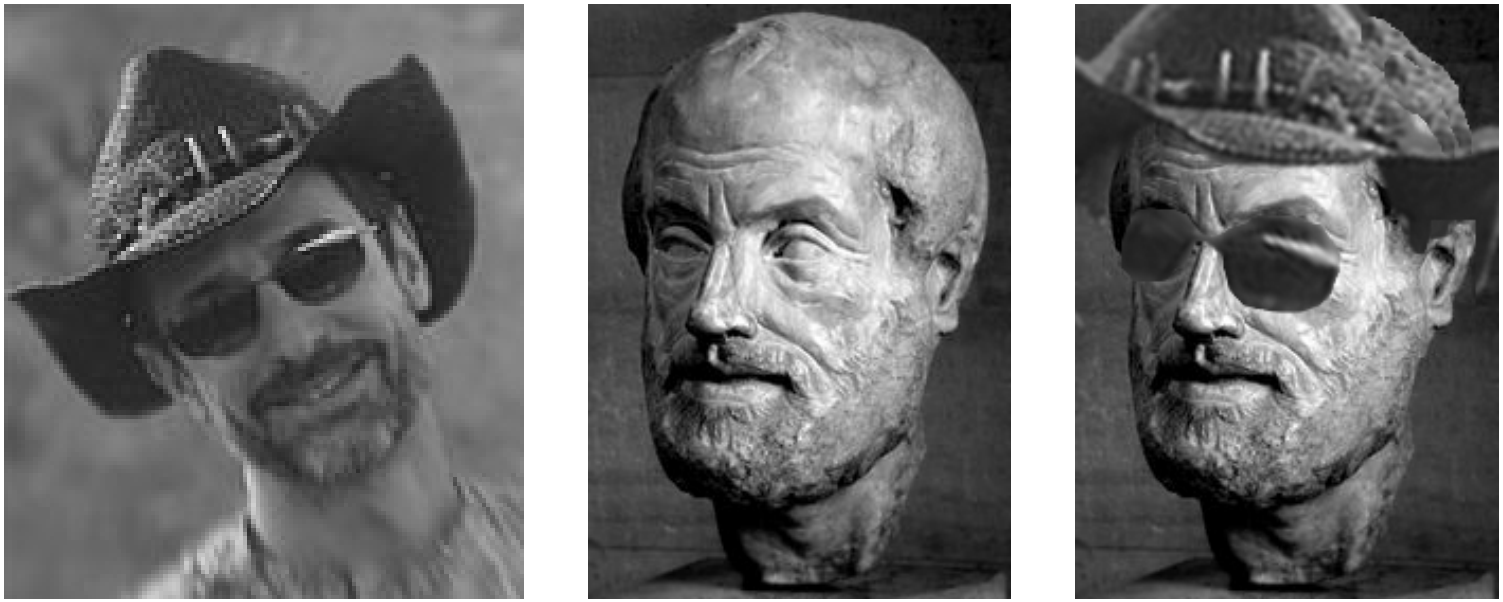


Figure: Tom and Aristotle...

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About this lesson: what is the Semantic Web

“ The Semantic Web is not a separate Web but an extension of the current one, in which information is given a well-defined meaning, better enabling computers and people to work in cooperation.”
Tim Berners-Lee, James Hendler and Ora Lassila; Scientific American, May 2001

- Automatic discovery of web resources and web services
- Automatic composition of web resources and web services
- Automatic execution and monitoring of web services

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Motivation: the Importance of Being Semantically Interoperable...

- Semantic interoperability, namely "*the ability of computer systems to exchange data with unambiguous, shared meaning*" (Wikipedia, `http://en.wikipedia.org/wiki/Semantic_interoperability`), is one of the most relevant and lively research fields of the last twentyfive years.
- The advent of ontologies in computer science in the early nineties, the settlement of Web Services in the beginning of the new millennium, the combination of both into Semantic Web Services, all witness the fervid activity of academia and industry for finding algorithms, languages, tools, and infrastructures aimed at making the "semantic interoperability dream" come true.

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Motivation: the Importance of Being Semantically Interoperable...

- The reason for the interest in this topic is easy to explain. In a very old paper, Jan Walker et al. assess the value of information exchange and interoperability in the domain of health care, and state that a fully standardised system supporting information exchange and interoperability could yield a net value of 77.8 billion dollars per year once implemented.
- Other much more recent papers and activities confirm the expected economic advantages in implementing solutions for semantic interoperability.

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Motivation: the Importance of Being Semantically Interoperable...

[The European Data Conference on Reference Data and Semantics 2023, https://op.europa.eu/en/web/endorse]
[The Value and Challenges of Semantic Inoperability in Healthcare 2018, https://www.usfhealthonline.com/resources/health-informatics/the-challenge-of-semantic-interoperability/]
[J. Walker, E. Pan, D. Johnston, J. Adler-Milstein, D. W. Bates and B. Middleton, *The Value of Healthcare Information Exchange and Interoperability*, Health Affairs, Web Exclusive, 19 January 2005]

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What is an ontology? - Tom Gruber's Definition, 2008

"In the context of computer and information sciences, an ontology defines a set of representational primitives with which to model a domain of knowledge or discourse.
The representational primitives are typically classes (or sets), attributes (or properties), and relationships (or relations among class members).
The definitions of the representational primitives include information about their meaning and constraints on their logically consistent application."

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What is an ontology? - Tom Gruber's Definition, 2008

"In the context of database systems, ontology can be viewed as a level of abstraction of data models, analogous to hierarchical and relational models, but intended for modeling knowledge about individuals, their attributes, and their relationships to other individuals.
Ontologies are typically specified in languages that allow abstraction away from data structures and implementation strategies; in practice, the languages of ontologies are closer in expressive power to first-order logic than languages used to model databases.
For this reason, ontologies are said to be at the "semantic" level, whereas database schema are models of data at the "logical" or "physical" level."

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What is an ontology? - Tom Gruber's Definition, 2008

"Due to their independence from lower level data models, ontologies are used for integrating heterogeneous databases, enabling interoperability among disparate systems, and specifying interfaces to independent, knowledge-based services. In the technology stack of the Semantic Web standards, ontologies are called out as an explicit layer. There are now standard languages and a variety of commercial and open source tools for creating and working with ontologies."

Tom Gruber, Ontology, to appear in the Encyclopedia of Database Systems, Ling Liu and M. Tamer Özsu (Eds.), Springer-Verlag, 2008.

`http://tomgruber.org/writing/ontology-definition-2007.htm`

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An Example

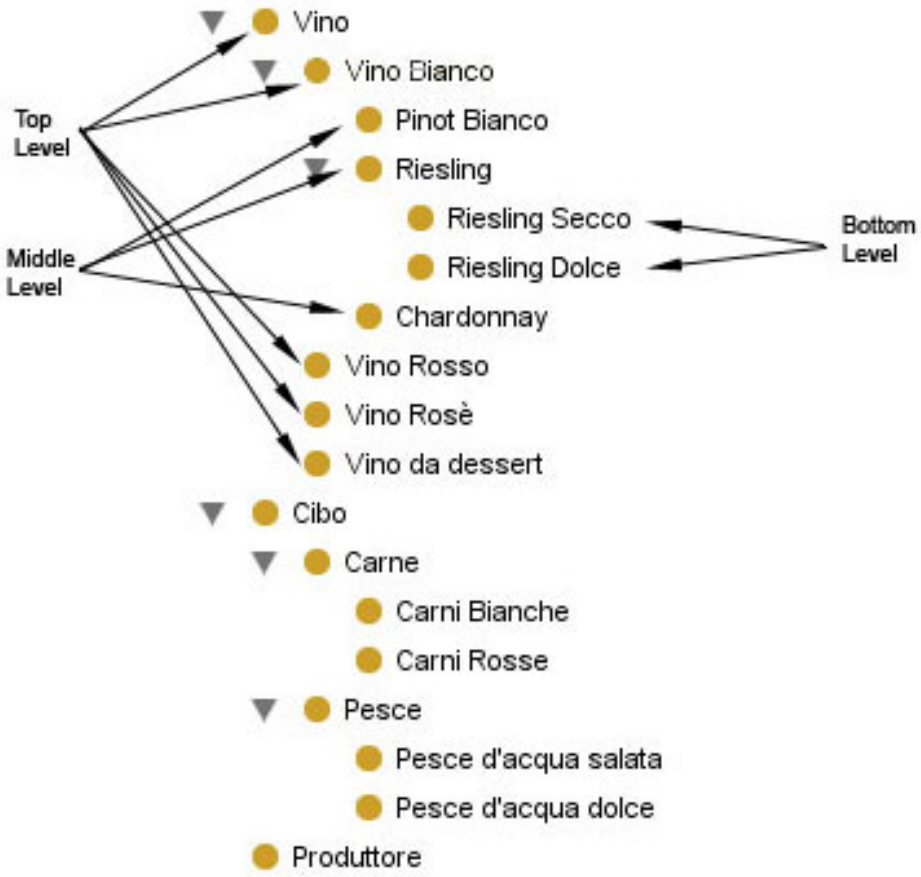


Figure: Wine Ontology

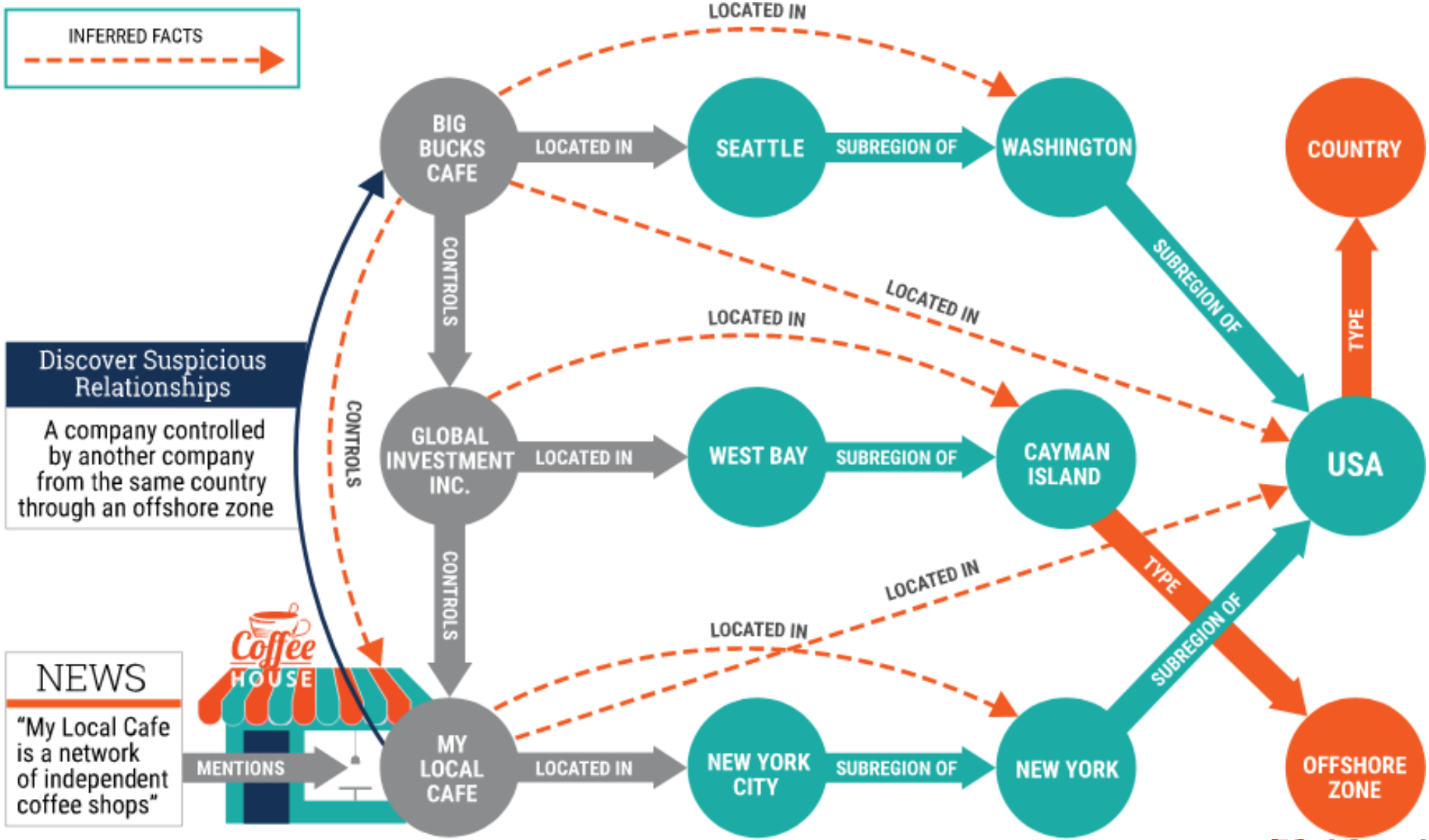
Ontologies and Databases

- **Purpose**
  Ontologies: Sharing, reuse, defining semantics.
  Databases: storing large amounts of structured data.
- **Expressive power of modelling languages**
  Ontologies: close first order logic; description logics.
  Databases: not as powerful as FOL.

Ontologies and Databases

- **Standards for making queries**
  Ontologies: no standards, few proposals, for example SPARQL.
  Databases: consolidated standards, for example SQL.
- **Management systems**
  Ontologies: Protégé, Jena (not commercial).
  Databases: Many solid, commercial tools and environments (Oracle, etc).

Knowledge Graphs



Knowledge Graphs

The heart of the knowledge graph is a knowledge model: a collection of interlinked descriptions of concepts, entities, relationships and events. Knowledge graphs put data in context via linking and semantic metadata and this way provide a framework for data integration, unification, analytics and sharing.

https://www.ontotext.com/knowledgehub/fundamentals/what-is-a-knowledge-graph/

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Methodology for ontology development

Natalya F. Noy and Deborah L. McGuinness, Ontology Development 101: A Guide to Creating Your First Ontology, Knowledge Systems Laboratory, March, 2001.
http://protege.stanford.edu/publications/ontology_development/ontology101-noy-mcguinness.html

- Determine the domain and scope of the ontology
- Consider reusing existing ontologies
- Enumerate important terms in the ontology
- Define the classes and the class hierarchy
- Define the properties of classes-slots
- Define the facets of the slots
- Create instances

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Languages Evolution

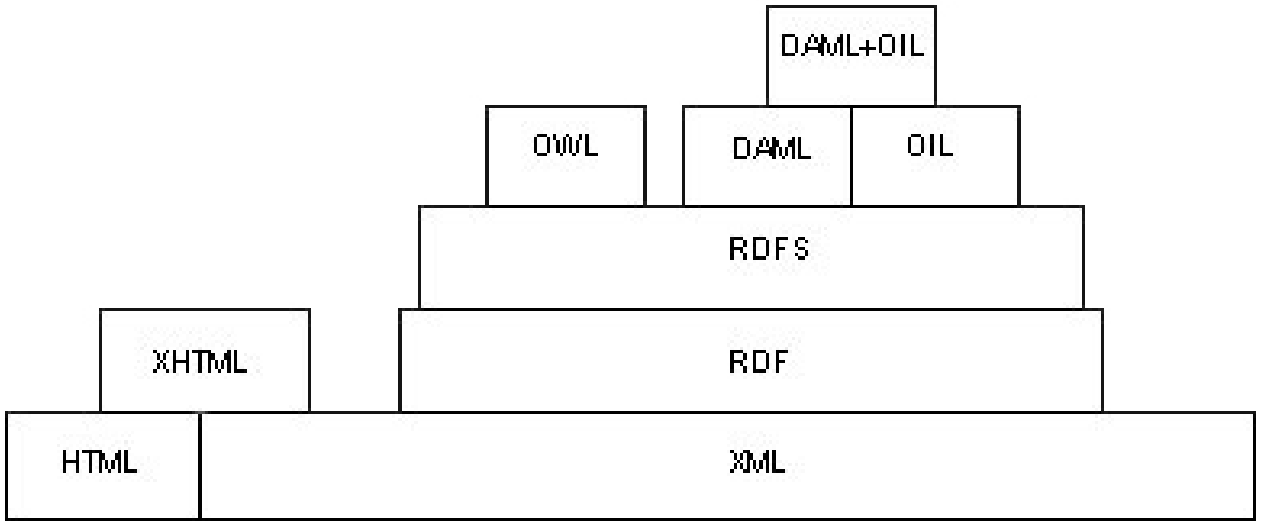


Figure: Languages Evolution

RDF - *RDF Model and Syntax & RDF Schema*

RDF is a W3C recommendation and provides a general, flexible method to decompose any knowledge into small pieces, called triples, with some rules about the semantics (meaning) of those pieces.

The RDF data model provides three object types:

- A resource may be either entire Web page, a part of it, a whole collections of pages, or an object that is not directly accessible via the Web;
- A property is a specific aspect, characteristics, attribute, or relation used to describe a resource.
- A statement is a triple consisting of two nodes and a connecting edge. These basic elements are all kinds of RDF resources

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Statement RDF

Basic Idea: triple (O, A, V) more frequently written as $A(O, V)$.

Meaning: an object O has an attribute A with value V.

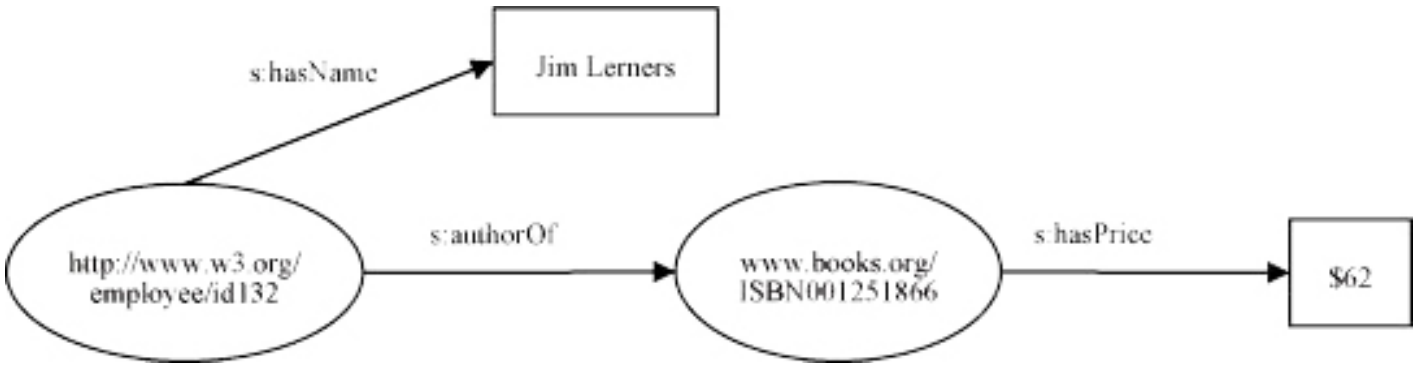


Figure: An RDF Graph

`http://www.w3.org/TR/REC-rdf-syntax/`

Example

```
<?xml version="1.0"?>

<RDF>
  <Description about="http://www.w3schools.com/rdf">
    <author>Jan Egil Refsnes</author>
    <homepage>http://www.w3schools.com</homepage>
  </Description>
</RDF>
```

RDF identifies things using Web identifiers (URIs), and describes resources with properties and property values.
Explanation of Resource, Property, and Property value:

- A Resource is anything that can have a URI, such as "http://www.w3schools.com/rdf"
- A Property is a Resource that has a name, such as "author" or "homepage"
- A Property value is the value of a Property, such as "Jan Egil Refsnes" or "http://www.w3schools.com" (note that a property value can be another resource)

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Example

The combination of a Resource, a Property, and a Property value forms a Statement (known as the subject, predicate and object of a Statement).
Let's look at some example statements to get a better understanding:
Statement: "The author of http://www.w3schools.com/rdf is Jan Egil Refsnes".

- The subject of the statement above is: http://www.w3schools.com/rdf
- The predicate is: author
- The object is: Jan Egil Refsnes

Statement: "The homepage of http://www.w3schools.com/rdf is http://www.w3schools.com".

- The subject of the statement above is: http://www.w3schools.com/rdf
- The predicate is: homepage
- The object is: http://www.w3schools.com

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RDF Schema

RDF schemas are used to define the structure of the metadata that are used to describe WWW resources (i.e. WWW pages or parts of WWW pages, referenced by an URL).

The RDF Schema Specification consists of some basic classes and properties, and can be extended by others to fit possibly any given domain.

Classes are arranged hierarchically, and the use of properties can be constrained to members of certain classes. The root of the class hierarchy is `rdfs:Resource`, `rdfs:Class` is subclass of `rdfs:Resource`.

Properties are defined by the `rdf:Property` class and can be seen as attributes, that are used to describe resources by assigning values to them. Properties are resources themselves.

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Example RDFS

```
<rdfs:Class rdf:ID="Teacher">
  <rdfs:comment>Teacher Class</rdfs:comment>
  <rdfs:subClassOf rdf:resource="#Person"/>
</rdfs:Class>

<rdfs:Class rdf:ID="Course">
  <rdfs:comment>Course Class</rdfs:comment>
  <rdfs:subClassOf rdf:resource="http://www.w3.org/1999/02/22-rdf-syntax-ns#Resource"/>
</rdfs:Class>

<rdf:Property rdf:ID="teacher">
   <rdfs:comment>Teacher of a course</rdfs:comment>
   <rdfs:domain rdf:resource="#Course"/>
   <rdfs:range rdf:resource="#Teacher"/>
</rdf:Property>

<rdf:Property rdf:ID="name">
   <rdfs:comment>Name of a Person or Course</rdfs:comment>
   <rdfs:domain rdf:resource="#Person"/>
   <rdfs:domain rdf:resource="#Course"/>
   <rdfs:range rdf:resource="http://www.w3.org/1999/02/22-rdf-syntax-ns#Literal"/>
</rdf:Property>
```

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OWL (Ontology Web Language)

OWL is an extension of RDF Schema, in the sense that OWL would use the RDF meaning of classes and properties (`rdfs:Class`, `rdfs:subClassOf`, etc), and would add language primitives to support a richer expressiveness.

- **OWL Full:** It uses all the OWL languages primitives. It also allows to combine these primitives in arbitrary ways with RDF and RDF Schema.
  The advantage of OWL Full is that it is fully upward compatible with RDF.
  The disadvantage of OWL Full is that the language has become so powerful as to be undecidable.

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OWL (Ontology Web Language

- **OWL DL:** It is a sublanguage of OWL Full which restricts the way in which the constructors from OWL and RDF can be used.
  The advantage of this is that it permits efficient reasoning support.
  The disadvantage is that we loose full compatibility with RDF.

- **OWL Lite:** An ever further restriction limits OWL DL to a subset of the language constructors.
  The advantage of this is a language that is both easier to grasp (for users) and easier to implement (for tool builders).
  The disadvantage is of course a restricted expressivity.

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OWL Syntax

```
<rdf:RDF
   xmlns:owl ="http://www.w3.org/2002/07/owl#"
   xmlns:rdf ="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
   xmlns:rdfs="http://www.w3.org/2000/01/rdf-schema#"
   xmlns:xsd ="http://www.w3.org/2001/XLMSchema#">

<owl:Ontology rdf:about="">
   <rdfs:comment>An example OWL ontology</rdfs:comment>
   <owl:priorVersion
      rdf:resource="http://www.mydomain.org/uni-ns-old"/>
   <owl:imports rdf:resource="http://www.mydomain.org/persons"/>
   <rdfs:label>University Ontology</rdfs:label>
</owl:Ontology>
```

`owl:imports` it includes other ontologies whose content is assumed to be part of the current document.

Also note that owl:imports is a transitive property: if ontology A imports ontology B, and ontology B imports ontology C, then ontology A also imports ontology C.

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Classes in OWL

Classes are defined using the element `owl:Class`

```
<owl:Class rdf:ID="Mammalia">
  <rdfs:subClassOf rdf:resource="#Animal"/>
</owl:Class>
```

For example We can add the information that this class is disjoint with the class `Insecta` through the element `owl:disjointWith`.

```
<owl:Class rdf:about="Mammalia">
  <owl:disjointWith rdf:resource="#Insecta"/>
</owl:Class>
```

There are two predefined classes: `owl:Thing` and `owl:Nothing`.

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Property in OWL

Two types of property:

- *Object properties* – link individuals to individuals.
- *Datatype properties* – link individuals to data values.

```
<owl:DatatypeProperty rdf:ID="age">
  <rdfs:range rdf:resource="http://www.w3.org/2001/XMLSchema
  #nonNegativeInteger"/>
  <owl:domain rdf:resource="#Person"/>
</owl:DatatypeProperty>
```

```
<owl:ObjectProperty rdf:ID="married">
  <owl:domain rdf:resource="#Person"/>
  <owl:range rdf:resource="#Person"/>
</owl:ObjectProperty>
```

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URLs

- RDF – `http://www.w3.org/RDF/`
- RDFS – `http://www.w3.org/TR/rdf-schema/`
- OWL – `http://www.w3.org/2004/OWL/`

Protégé

Protégé is a free, open source ontology editor and knowledge-base framework.
Developed by the Stanford University, Protégé is based on Java, is extensible, and provides a plug-and-play environment that makes it a flexible base for rapid prototyping and application development.

The Protégé platform supports two main ways of modeling ontologies:

- The Protégé-Frames editor enables users to build and populate ontologies that are frame-based.
- The Protégé-OWL editor enables users to build ontologies for the Semantic Web, in particular in the W3C's.

For more details: `http://protege.stanford.edu/`

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The Semantic Web is not a separate Web but an extension of the current one, in which information is given well-defined meaning, better enabling computers and people to work in cooperation.

The Semantic Web will bring structure to the meaningful content of Web pages, creating an environment where software agents roaming from page to page can readily carry out sophisticated tasks for users.



Tim Berners-Lee, James Hendler and Ora Lassila; Scientific American, May 2001.

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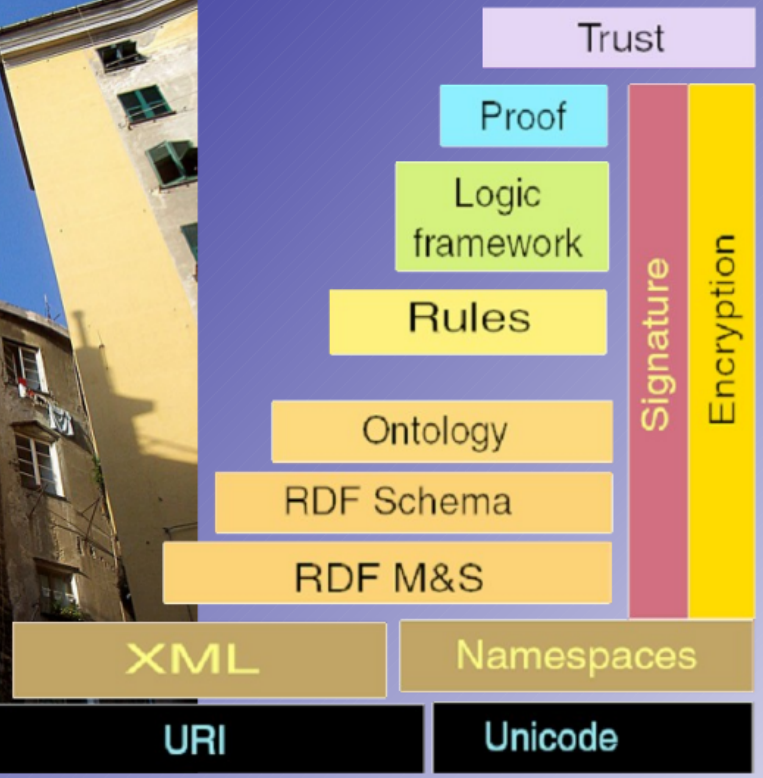


Figure: Tim, James, Ora

The Semantic Web Stack(s)

[Embriaci Tower, Genova]



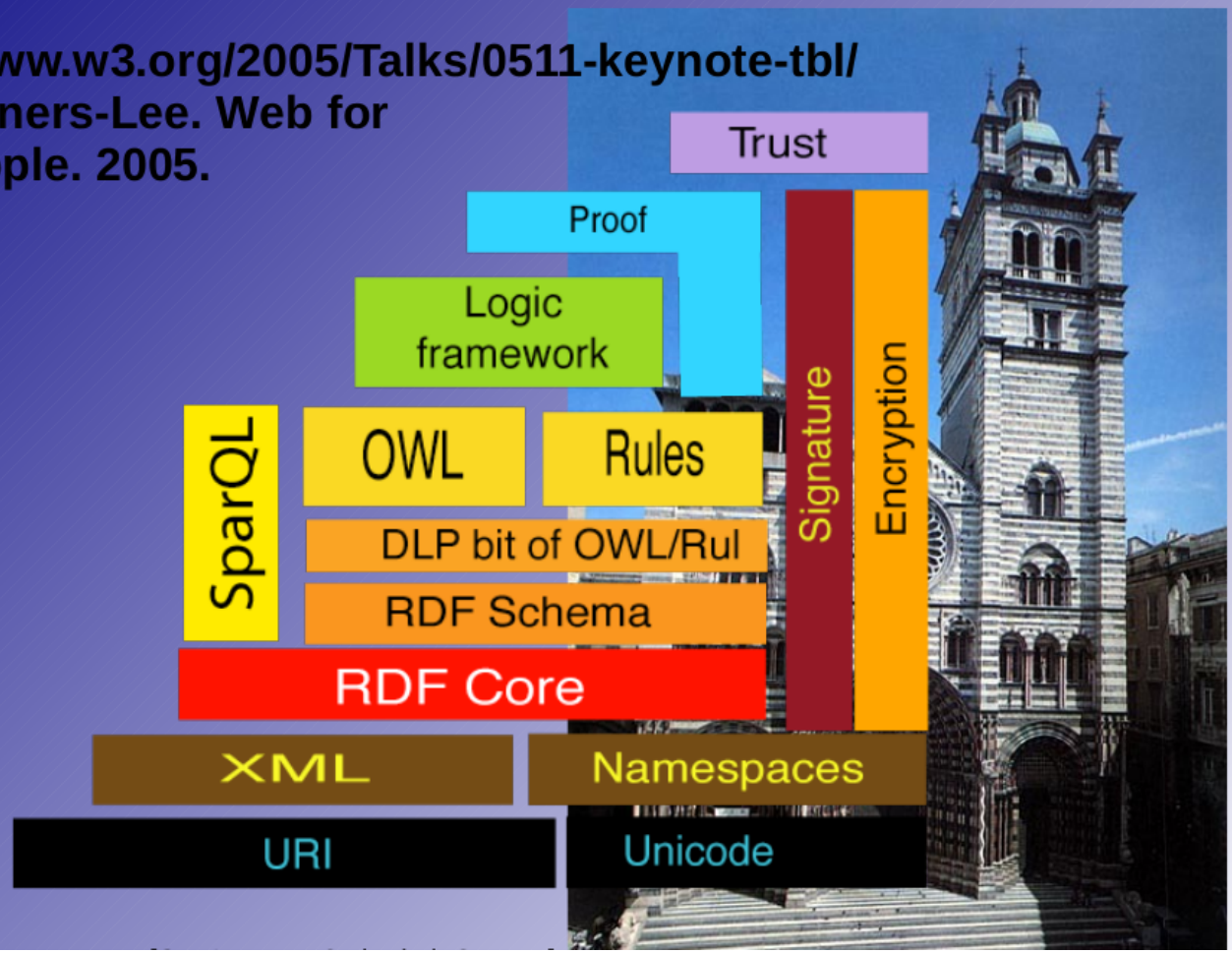


Tim Berners-Lee. WWW past & future, 2003. Available at http://www.w3.org/2003/Talks/0922-rsoc-tbl/

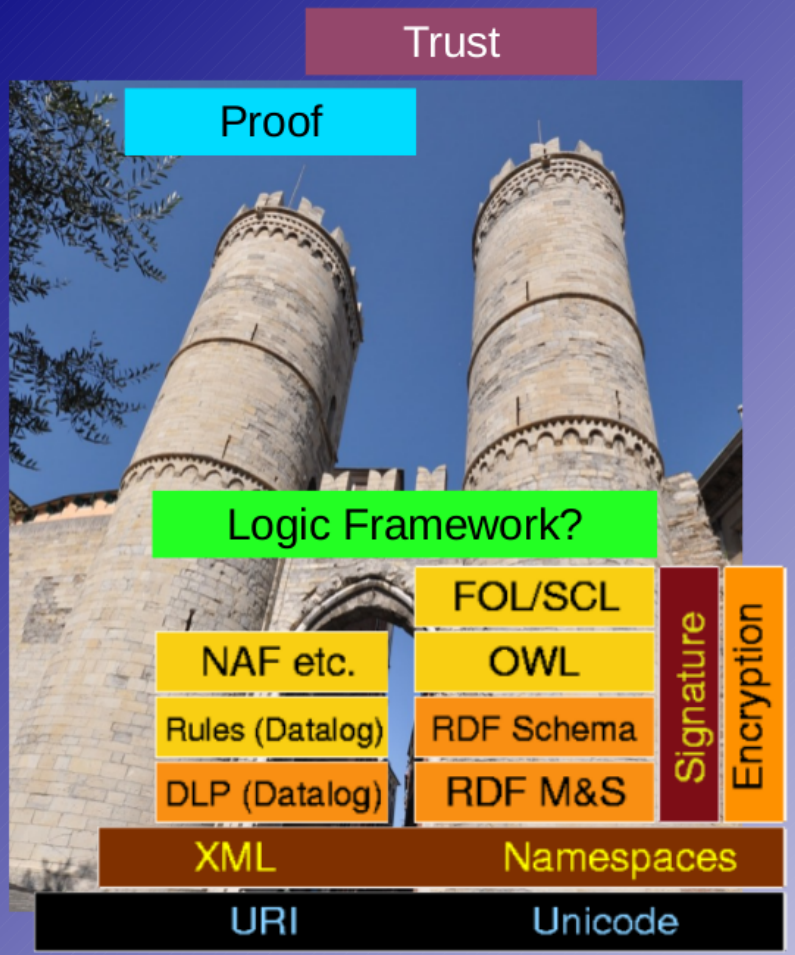
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The Semantic Web Stack(s)

http://www.w3.org/2005/Talks/0511-keynote-tbl/
Tim Berners-Lee. Web for real people. 2005.



The Semantic Web Stack(s)



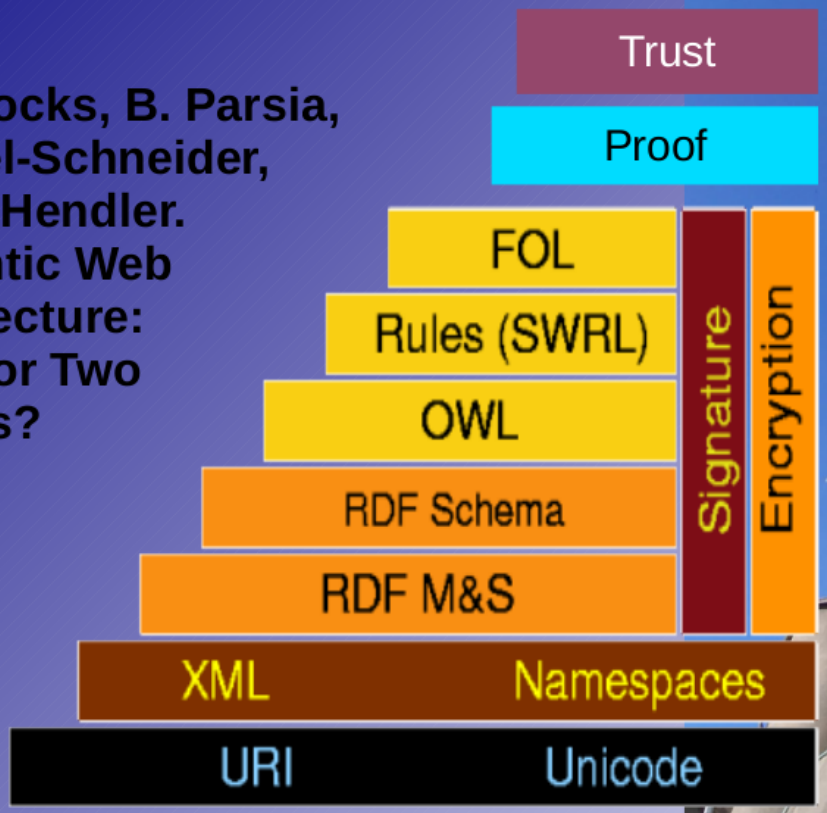
[Soprana Gate, Genova]

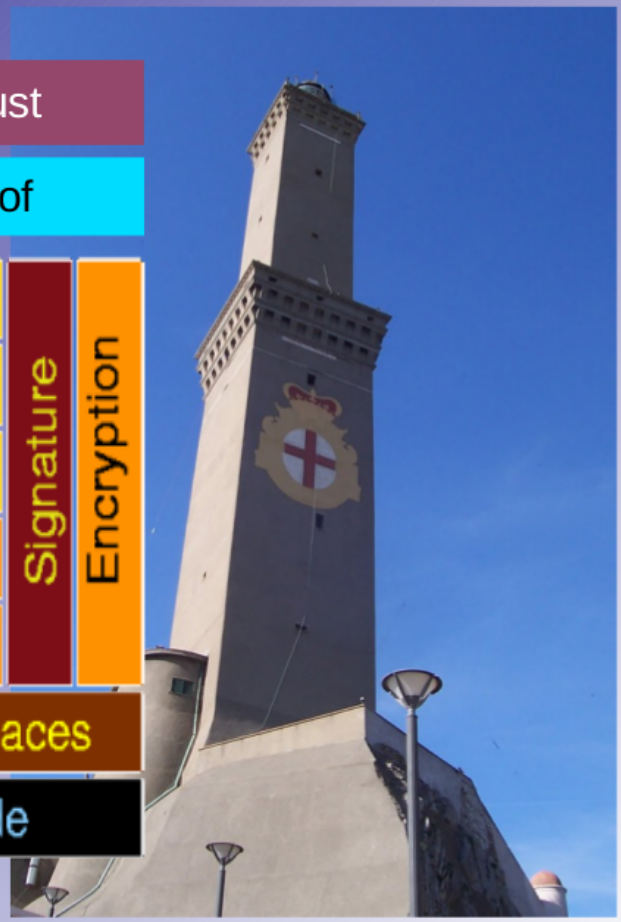
Ian Horrocks, Bijan Parsia, Peter Patel-Schneider, and James Hendler. Semantic Web Architecture: Stack or Two Towers? 3rd Int. conf. on Principles and Practice of Semantic Web Reasoning, 2005.

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The Semantic Web Stack(s)

I. Horrocks, B. Parsia, P. Patel-Schneider, and J. Hendler. Semantic Web Architecture: Stack or Two Towers? 2005.





[The “Lanterna” lighthouse, Genova]

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But, what is the Semantic Web really all about?



Linked Data (or Web of Data) – Get datasets rendered into RDF, use very simple ontologies (really more like thesauri or vocabularies than AI languages) to provide terms that can be used for linking and/or describing the data.

Web 3.0 – sometimes used as a synonym for Semantic Web, this term really grew up as a way of explaining how semantics (especially the simple ontologies we've been discussing here) can be integrated with Web 2.0 technologies to best effect.

Semantic Web – here's the tough one - some people use it to mean all of the above (I tend to).

James Hendler, http://blogs.nature.com/jhendler/2009/06/16/what-is-the-semantic-web-really-all-about

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Web 4.0

Web 4.0 services will be autonomous, proactive, content-exploring, self-learning, collaborative, and content-generating agents based on fully matured semantic and reasoning technologies as well as AI. They will support adaptive content presentation that will use the Web database via an intelligent agent. Examples might be services interacting with sensors and implants, natural-language services, or virtual reality services.

Source: `https://www.igi-global.com/dictionary/overview-differentiation-evolutionary-steps-web/35103`

Web 5.0

Control over identity: Web 5.0 will include identity management for users. Users will be able to store the credentials of all other platforms on a decentralized platform. This storage platform will only be controlled by the users, and the internet will not collect any information about the users' identities.
Data ownership: Using complete decentralization, this iteration of the Internet will provide users with complete data ownership. Users can choose what information to share with companies and what not to share. Users will have their own blockchain participation in order to store their data securely in a single location.

Source: `https://www.lcx.com/introduction-to-web-5-0/`

Actually... people seem no longer very interested in numbering the web versions!

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